

Homework 1 (exam)

1) OP: AC OP opened

$$V_{CE} - I_{C1}R_E + V_{BE} + I_{B1}R_B = 0$$

$$V_{CE} + V_{BE} = I_C = -0.746 \text{ mA}$$

$$R_E(1+\beta) + R_B$$

$$V_{BE} = V_{CE} - I_{C1}R_E + V_{BE} + I_{B1}R_B = 0$$

$$V_{CE} = -V_{CE} - I_{C1}R_E - I_{B1}(1+\beta)R_E = -6.65 \text{ V}$$

$$V_{B1} = -I_{C1}R_E = +7.46 \text{ V}$$

$$V_{CE} - I_{C1}R_D - V_{DS} - I_{D1}R_S = 0$$

$$V_{DS} = V_{CE} - I_{C1}R_D \rightarrow I_D = \frac{V_{CE} - V_{DS}}{R_S}$$

$$I_D = I_{D1} \cdot (1 - V_{DS})^2$$

$$\frac{V_{CE} - V_{DS}}{R_S} = I_{D1} + \frac{I_{D1} V_{DS}^2}{V_{T^2}} - \frac{2 I_{D1} V_{DS}}{V_T} \rightarrow \frac{I_{D1} V_{DS}^2}{V_{T^2}} + V_{DS} \left(\frac{1}{R_S} - \frac{2 I_{D1}}{V_T} \right) + I_{D1} - \frac{V_{CE}}{R_S} = 0$$

$$(5 \cdot 10^{-5}) + V_{DS} (8 \cdot 10^{-4}) + I_{D1} - (2.26 \cdot 10^{-3}) = 0$$

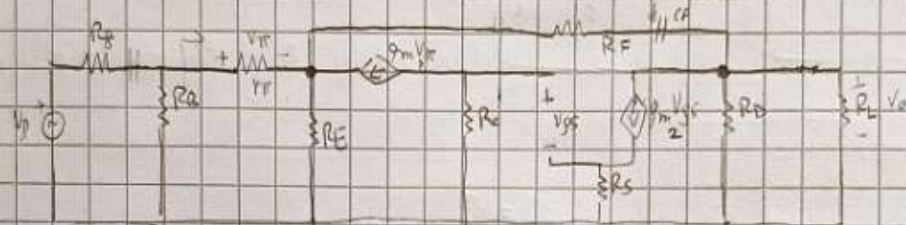
$$V_{DS} = 6.65 \text{ V} > V_{CE}$$

$$I_D = 9.9 \cdot 10^{-4} = 1 \text{ mA}$$

$$V_{DS} = V_{CE} - I_D(R_D + R_S) = 7.2 \text{ V}$$

$$2) A_v = \frac{V_o}{V_g}$$

starting Pice caps



Feedback topology

Voltage mixing at input $\rightarrow$ single loop $\rightarrow$ voltage divider

Voltage sampling $\rightarrow$ single node, short structure $\rightarrow$ sensing voltage

voltage amplifier $A_v = \frac{V_o}{V_g}$



beta network

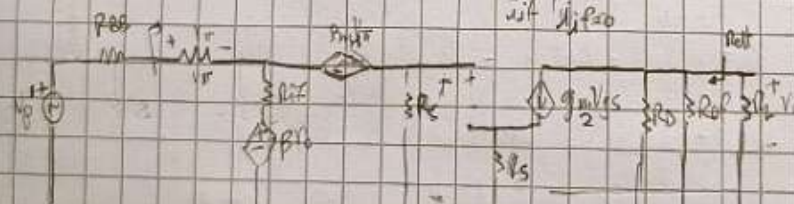


$$R_{if} = \frac{V_{if}}{I_{if}} = \frac{V_{CE}}{I_{C1}} = 776 \Omega$$

$$R_{of} = \frac{V_{of}}{I_{of}} = R_S + R_E = 3.4 \text{ k}\Omega$$

$$\beta = \frac{V_{CE}}{V_{DS}} = \frac{R_E}{R_D + R_E} = 0.353$$

β is negligible



Theorem at input:

$$V_g' = V_g \frac{R_B}{R_B + R_S}$$

$$R_{BB} = R_B \parallel R_S = 2.95 \text{ k}\Omega$$

$$g_{m1} = \frac{I_{D1}}{V_T} = 0.098 \text{ A/V}$$

$$g_{m2} = \frac{2 I_D}{V_{GS} - V_T} = 4.5 \cdot 10^{-4} \text{ A/V}$$

$$V_{T1} = \frac{R_{BB}}{g_{m1}} = 3.36 \text{ k}\Omega$$

$$A = \frac{V_o}{V_g'} = \alpha_i \cdot \alpha_{cs} \cdot \alpha_{cs} = 7.13$$

$$A_f = \frac{A_v}{1 + \beta A_v} = \frac{V_o}{V_g'} = 2.055$$

$$\alpha_{cs1} = \frac{-g_{m2} V_{DS} (R_D \parallel R_{of} \parallel R_L)}{V_{DS} + V_{GS} \frac{R_D \parallel R_{of} \parallel R_L}{R_S}} = \frac{-g_{m2} (R_D \parallel R_{of} \parallel R_L)}{1 + g_{m2} R_S} = -0.688$$

$$A = \frac{V_o}{V_g} = A_f \cdot \frac{V_g'}{V_g} = A_f \cdot K_i = A_f \cdot \frac{R_B}{R_B + R_S} = 2.044$$

$$\alpha_{cs2} = \frac{-g_{m1} V_{CE} R_E}{V_{CE} + V_{BE} + g_{m1} V_{CE} R_E} = \frac{-\beta_0 R_E}{V_{T1} + (1 + \beta_0) R_{if}} = -12.23$$

$$R_{out} = R_D \parallel R_{of} \parallel R_L = 2.216 \text{ k}\Omega$$

$$R_{of} = \frac{R_{out}}{1 + \beta A_v} = 608 \Omega$$

$$R_{in,cs2} = V_{T1} + (1 + \beta_0) R_{if} = 81.76 \text{ k}\Omega$$

$$R_{of} = \frac{1}{\frac{1}{R_{of}} + \frac{1}{R_L}} = 612 \Omega$$

$$\alpha_{cs1} = \frac{R_{in,cs2}}{R_{BB} + R_{in,cs2}} = 0.89$$

$$f_L = \frac{1}{2\pi} \sum \frac{1}{R_i C_i} = \frac{1}{2\pi} \left(\frac{1}{R_g R_g^{sc}} + \frac{1}{L R^{sc}} + \frac{1}{C F R^{sc}} \right) = \frac{1}{2\pi} \left(\frac{1}{10\text{F} \cdot 284.7\text{K}\Omega} + \frac{1}{1\text{F} \cdot 100\text{G}\Omega} + \frac{1}{1\text{F} \cdot 368\text{K}\Omega} \right)$$

$$= 75.7 \text{ Hz} \quad \checkmark$$

$$R^{sc} = R_L + R_{out} = 100.62\text{K}\Omega$$

$$R_{in}^A = R_{BA} + D_{in}^{A2} = 91.7\text{K}\Omega$$

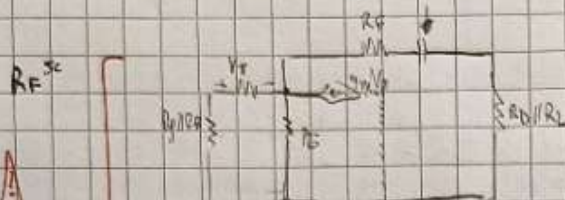
$$R_{in}^F = R_{in}^A (1 + \beta A_{in}) = 334.15\text{K}\Omega$$

$$R_{in} = (R_{in}^F - R_{BA}) \parallel R_B = 276.7\text{K}\Omega$$

$$R_g^{sc} = R_g + R_{in} = 284.7\text{K}\Omega$$

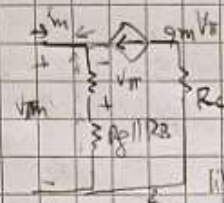
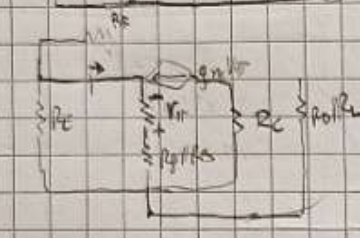
Can be applied because ω is very small

exact value: correct but a lot of calculations, easier with feedback theory but careful when it can be applied



$$R_F^{sc} = R_D \parallel R + R_F + R_E \parallel \left(\frac{R_F + R \parallel R_B}{1 + \beta_0} \right) = 3.68\text{K}\Omega$$

looking inside a common base



$$V_{in} = \frac{-V_{pi} - \frac{V_{pi} (R_F \parallel R_B)}{R_E}}{-\frac{V_{pi}}{R_E} - \frac{g_m V_{pi}}{R_E}} = \frac{R_E + R_F \parallel R_B}{(1 + \beta_0)}$$

like entering from the emitter of a common base

Ex 2

$$R_1 = 18\text{K}\Omega$$

$$R_2 = 56\text{K}\Omega$$

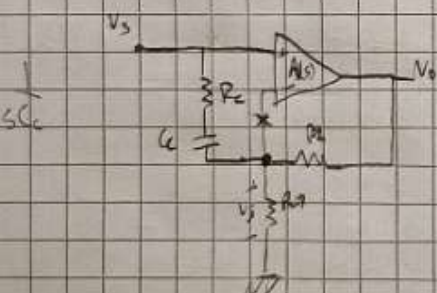
$$A(s) = \frac{A_0}{\left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right)}$$

$$A_0 = 10^5$$

$$\omega_{p1} = 100 \text{ rad/s}$$

$$\omega_{p2} = 5 \cdot 10^5 \text{ rad/s}$$

direct loop inspection!



1) phase margin with $R_2 = 0$

$$V_i = V_s - \frac{V_o R_1}{R_1 + R_2}$$

$$V_o = V_i A(s) = A(s) V_s - \frac{V_o A(s) R_1}{R_1 + R_2}$$

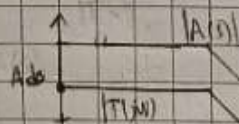
$$V_o \left(1 + \frac{A(s) R_1}{R_1 + R_2} \right) = A(s) V_s$$

$$\text{closed loop } \frac{V_o}{V_s} = \frac{A(s) (R_1 + R_2)}{(R_1 + R_2) + A(s) R_1} \xrightarrow{A(s) \rightarrow \infty} \frac{R_1 + R_2}{R_1} = 1 + \frac{R_2}{R_1} \quad (\text{Korn's value})$$

$$T = \beta A(s) = \frac{R_1}{R_1 + R_2} A(s)$$

$$\beta T = \frac{V_i}{V_o} = \frac{R_1}{R_1 + R_2}$$

$$A_0 = 88\text{dB}$$

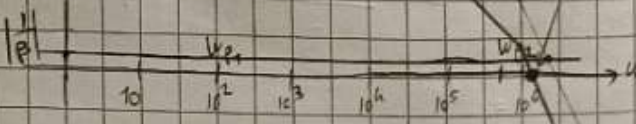


$$T = \frac{A_0}{\left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right)} \quad A_0 = \frac{R_1}{R_1 + R_2} A_0$$

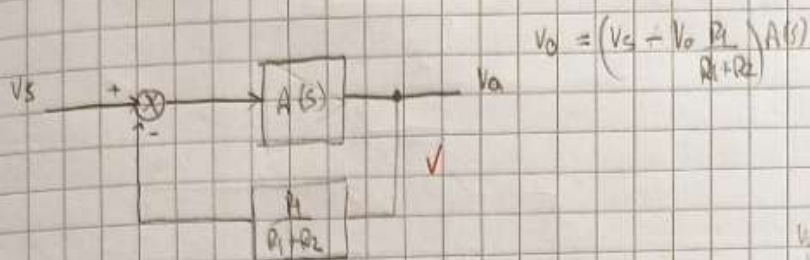
Since we are crossing 20dB

$$\angle A(j\omega) = -160^\circ \rightarrow \text{aw } 20^\circ \text{ PM}$$

we are crossing 0dB with +40dB/dec so the phase margin



2) block diagram representation of the amplifier (direct loop inspection)



$$V_o = \left(V_s - V_o \frac{R_1}{R_1 + R_2} \right) A(s) \quad \checkmark$$

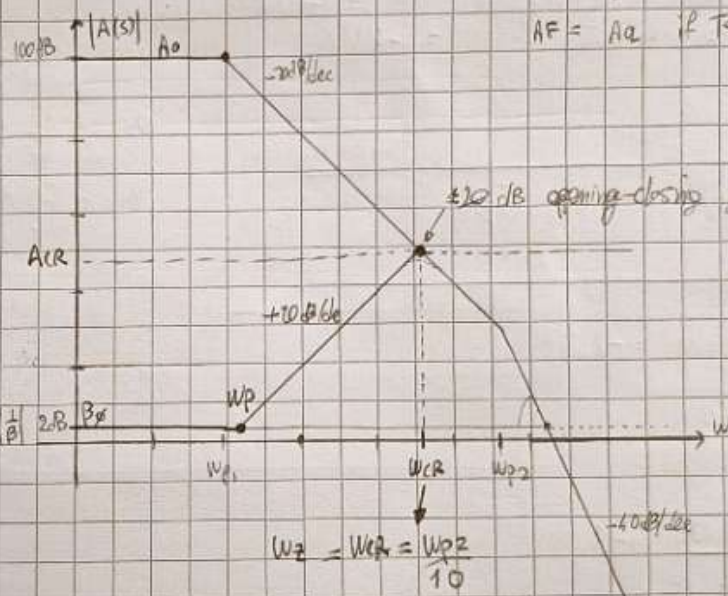
3) if $T \gg 1$ $AF = \frac{A(s)}{1 + A(s) \frac{R_1}{R_1 + R_2}} = \frac{1}{\beta(s)} = \left[\frac{R_1}{(R_1 + R_2)} \cdot \frac{1 + sC_c R_c}{1 + sC_c \left[\frac{(R_c + R_1)R_2 + R_c R_1}{R_1 + R_2} \right]} \right]^{-1}$

$$\beta(s) = \frac{V_o}{V_s} = \frac{\left(R_c + \frac{1}{sC_c} \right) \parallel R_1}{R_2 + R_1 \parallel \left(R_c + \frac{1}{sC_c} \right)} = \frac{R_1 (1 + sC_c R_c)}{R_2 + \frac{R_1 (1 + sC_c R_c)}{1 + sC_c (R_c + R_1)}} = \frac{R_1 (1 + sC_c R_c)}{R_2 (1 + sC_c (R_c + R_1)) + R_1 (1 + sC_c R_c)}$$

$$\frac{sC_c R_c + 1}{sC_c} \parallel R_1 = \frac{R_1 (1 + sC_c R_c)}{sC_c \cdot (sC_c R_c + 1) \times R_1} = \frac{R_1 (1 + sC_c R_c)}{(1 + sC_c (R_c + R_1))}$$

$$AF = \frac{1}{\beta} \quad \text{if } T \gg 1$$

$$AF = A_c \quad \text{if } T \ll 1$$



$$\beta_c(0) = \frac{R_1}{R_1 + R_2} \cdot \left(\frac{1 + s/w_c}{1 + s/w_p} \right)$$

$$\frac{1}{\beta_c(s)} = \frac{R_1 + R_2}{R_1} \cdot \left(\frac{1 + s/w_p}{1 + s/w_c} \right)$$

$$w_{cR} = 5 \cdot 10^4 \text{ rad/s} = w_c$$

$$w_z = \frac{1}{C_c R_c}$$

$$\left\{ \begin{array}{l} A_0 w_{p1} = A_{cr} \cdot w_{cR} \\ \frac{1}{\beta_c} \cdot \frac{1}{w_p} = \frac{A_{cr}}{w_{cR}} \end{array} \right\} \quad \left\{ \begin{array}{l} A_{cr} = \frac{A_0 w_{p1}}{w_{cR}} \\ w_p = \frac{1}{\beta_c A_0 w_{p1}} \cdot w_{cR} = \frac{w_{cR}^2}{\beta_c A_0 w_{p1}} = 1028 \text{ rad/s} \approx 10^3 \text{ rad/s} \end{array} \right.$$

$$\left\{ \begin{array}{l} w_z = \frac{1}{C_c R_c} \\ w_p = \frac{1}{C_c \left[\frac{(R_c + R_1)R_2 + R_c R_1}{R_1 + R_2} \right]} \end{array} \right\} \quad \left\{ \begin{array}{l} w_z = \frac{w_p \left[\frac{(R_c + R_1)R_2 + R_c R_1}{R_c (R_1 + R_2)} \right]}{R_c (R_1 + R_2)} \Rightarrow \frac{w_z}{w_p} = \frac{R_c (R_2 + R_1) + R_1 R_2}{R_c (R_1 + R_2)} \\ \frac{w_p}{w_z} \cdot R_c (R_1 + R_2) = R_c (R_2 + R_1) + R_1 R_2 \\ R_c = \frac{R_1 R_2}{\left(\frac{w_z}{w_p} - 1 \right) (R_1 + R_2)} = 285 \Omega \end{array} \right.$$

$$C_c = \frac{1}{w_z R_c} = 67 \text{ nF}$$

$$PM: \angle T(jw) = \angle A(s) \cdot \beta_c(s) = -140^\circ \quad \downarrow +40^\circ = PM$$